# CHAPTER 1 — INTRODUCTION

## CONTENTS — MAINTENANCE PROCEDURES


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# SEARCH AND RESCUE SUPPLEMENT FOR BELL MODEL 412 HELICOPTER

## 1-1. DESCRIPTION.

This supplement contains information and instructions for maintenance of instrument, electrical, avionic, and flight control systems/components used in Search and Rescue (SAR) configured Bell Model 412 helicopters.

Supplemental electrical and avionic systems in SAR configured Model 412 helicopters consists of Digital Automatic Flight Control system (DAFCS), Electronic Flight Instrument System (EFIS), radar altimeter navigation system and the weather radar system. Other customizing avionic equipment installed includes Doppler velocity sensor, DATA NAV III, and navigation system (GNS-X).

The DDAFCS consists of dual/autopilot flight director computers, mode selector, autopilot controller, air data sensors and servos. It also comprises the standard flight controls plus added (DAFCS) (SAR) components in the collective, cyclic, and directional control systems. Mechanical linkage systems, actuated by electrical actuators and conventional helicopter controls are used to control flight attitude, altitude and direction.

The system is fully coupled, four axis (pitch, roll, yaw, and collective) system with SAR modes combining autopilot and flight director functions. The system also provides auto trim, heading hold, automatic turn coordination, and auto level.

The flight director computers provide the pilot and copilot with steering information to fly the helicopter manually or the information can be coupled to the autopilot for fully automatic flight path control. SAR specific modes include, radio altitude hold, velocity hold, mark-on-target, approach 1, approach 2, and climb is also coupled to the autopilots.

The EFIS is comprised of pilot and copilot electronic displays (electronic attitude director indicator (EADI) and electronic horizontal situation indicator (EHSI)), symbol generators, display controllers, and remote instrument controller.

The EFIS displays pitch and roll attitude, heading, course orientation, source annunciators, doppler velocities, weather presentations, navigation waypoints, and flightpaths.

The radio altimeter system consists of a receiver/transmitter, transmit and receive antennas, and an indicator. The receiver/transmitter provides radio altitude to the indicator, EFIS, and DAFCS from 0 to 2500 feet. The indicator displays radio altitude from 0 to 1500 feet.

The weather radar system consists of a receiver/transmitter, antenna, indicator, and controller. The system is a high resolution, high power airborne weather/ground-mapping radar optimized for over-water search and rescue missions.

Refer to chapters 97 and 98 for avionic and electrical wiring diagrams of SAR systems installed.

# ABBREVIATIONS

## 1-2. ABBREVIATIONS.

Abbreviations used in this manual are defined below.

| Abbreviation | Equivalent |
|---|---|
| ACTR | ACTUATOR |
| ADF | AUTOMATIC DIRECTION FINDER |
| ADI | ATTITUDE DIRECTOR INDICATOR |
| ADS | AIR DATA SENSOR |
| AFCS | AUTOMATIC FLIGHT CONTROL SYSTEM |
| ALT | ALTITUDE |
| ALTE | ALTITUDE ERROR |
| ALTM | ALTIMETER |
| ANN | ANNUNCIATOR |
| ANT | ANTENNA |
| AOG | AIRCRAFT ON-GROUND |
| AOSS | AFTER OVER STATION SENSOR |
| AP | AUTOPILOT |
| API | ACTUATOR POSITION INDICATOR |
| APR | APPROACH |
| APR HOV | APPROACH TO HOVER |
| AS, A/S | AIRSPEED |
| ATT | ATTITUDE |
| AZ | AZIMUTH |
| BC | BACK COURSE |
| BCN | BEACON |
| BRG | BEARING |
| CAP | CAPTURE |
| CBB | COLLECTIVE BAR BIAS |
| CE | COURSE ERROR |
| CLR DLY | CLEARANCE DELIVERY |
| CLTV, COLL | COLLECTIVE |
| CMD | COMMAND |
| CMT | CONTROL MOTION TRANSDUCER |
| COD | COLLECTIVE OUT-OF-DETENT |
| CONT | CONTROLLER |
| CPL | COUPLE |
| CRS | COURSE |
| CX | CONTROL TRANSFORMER |
| DAFCS | DIGITAL AUTOMATIC FLIGHT CONTOL SYSTEM |
| DCPL | DECOUPLE |
| DEC, DECR | DECREASE |
| DEG | DEGREE(S) |
| DEV | DEVIATION |
| DFCS | DIGITAL FLIGHT CONTROL SYSTEM |
| DG | DIRECTIONAL GYRO |
| DME | DISTANCE MEASURING EQUIPMENT |

| Abbreviation | Equivalent |
|---|---|
| DOPP | DOPPLER |
| DTRK | DESIRED TRACK |
| EFIS | ELECTRONIC FLIGHT INSTRUMENT SYSTEM |
| ENC | ENCODE/ENCODING |
| ERR | ERROR |
| EXC | EXCITATION |
| FD, F/D | FLIGHT DIRECTOR |
| FDBK | FEEDBACK |
| FTR | FORCE TRIM RELEASE |
| GA, G/A | GO-AROUND |
| GEN | GENERATOR |
| GMAP | GROUND MAP |
| GS, G/S | GLIDESLOPE |
| GP | GLIDEPATH |
| GSPD | GROUNDSPEED |
| HDG | HEADING |
| HSI | HORIZONTAL SITUATION INDICATOR |
| HOV | HOVER |
| IAS | INDICATED AIRSPEED |
| IASE | INDICATED AIRSPEED ERROR |
| IFR | INSTRUMENT FLIGHT RULES |
| ILS | INSTRUMENT LANDING SYSTEM |
| INC, INCR | INCREASE |
| IND | INDICATOR |
| INV | INVERTER |
| IVV | INSTANTANEOUS VERTICAL VELOCITY |
| KT | KNOT(S) |
| LBS | LATERAL BEAM SENSOR |
| LOC | LOCALIZER |
| LSS | LIGHTING SENSOR SYSTEM |
| LVC | LINE VOLTAGE COMPENSATOR |
| LVDT | LINEAR VARIABLE DIFFERENTIAL TRANSFORMER |
| MAG | MAGNETIC |
| MLS | MICROWAVE LANDING SYSTEM |
| MOT | MARK ON TARGET |
| NAV | NAVIGATION |
| NOC | NAV ON-COURSE |
| OSS | OVER STATION SENSOR |
| PBB | PITCH BAR BIAS |
| PCPL | PITCH COUPLE |
| PGE | PAGE |
| P/O | PART OF |
| POD | PITCH OUT-OF-DETENT |
| POSN | POSITION |
| PTT | PUSH-TO-TALK |
| PWM | PULSEWIDTH MODULATED |
| PWR | POWER |
| RA, R/A | RADIO ALTIMETER |
| RBB | ROLL BAR BIAS |
| RCPL | ROLL COUPLE |
| RCT or REACT | RAIN ECHO ATTENUATION COMPENSATION TECHNIQUE |

| Abbreviation | Equivalent |
|---|---|
| RCVR | RECEIVER |
| RNG | RANGE |
| ROD | ROLL OUT-OF-DETENT |
| RT, R/T | RECEIVER/TRANSMITTER |
| SAS | STABILITY AUGMENTATION SYSTEM |
| SBY | STANDBY |
| SCT | SECTOR |
| SEC | SECOND |
| SEL | SELECT |
| SQ | SQUELCH |
| SYNC | SYNCHRONIZATION |
| SYS | SYSTEM |
| TGT | TARGET |
| TRB or TURB | TURBULENCE |
| TTL | TUNED TO LOCALIZER |
| TX | TRANSMIT |
| VAPR | VOR APPROACH, ALSO VOR APR |
| VBS | VERTICAL BEAM SENSOR |
| VG | VERTICAL GYRO |
| VLD | VALID |
| VOR | VHF OMNI RANGE |
| VOR APP | VOR APPROACH |
| VS, V/S | VERTICAL SPEED |
| VSE | VERTICAL SPEED ERROR |
| VSI | VERTICAL SPEED INDICATOR |
| WC | WEATHER CONTROLLER |
| WI | WEATHER INDICATOR |
| W/L | WINGS LEVEL |
| WO, W/O | WASHOUT |
| WX | WEATHER RADAR |
| XMTR | TRANSMITTER |
| YOD | YAW OUT-OF-DETENT |